Linear Approximation and L'Hôpital's Rule

ANSWER KEY

Tangent line approximation

1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x)=\sqrt{x}$, $f(4)=2$, and $f'(4)=\frac{1}{4}$.
$L(x)=f(4)+f'(4)(x-4)=2+\frac{1}{4}(x-4)$. $L(4.1)=2+\frac{1}{4}(0.1)=2.025$.

2 Use linear approximation to estimate $(1.02)^5$, using $f(x)=x^5$ near $x=1$.
$f(1)=1$, $f'(x)=5x^4$, $f'(1)=5$. $L(x)=1+5(x-1)$. $L(1.02)=1+5(0.02)=1.1$.

3 Find the linearization $L(x)$ of $f(x)=\sin x$ at $x=0$, and use it to approximate $\sin(0.1)$.
$f(0)=0$, $f'(0)=\cos 0=1$: $L(x)=x$. $\sin(0.1)\approx L(0.1)=0.1$.

Error and over/underestimation

4 For $f(x)=\sqrt{x}$ (concave down for $x>0$), explain whether the linear approximation at $x=4$ overestimates or underestimates $f(4.1)$.
Since f is concave down, its graph lies below any tangent line, so the tangent line (linear approximation) OVERESTIMATES the true value of $f(4.1)$.

5 For $f(x)=x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.
Since f is concave up, its graph lies above any tangent line, so the tangent line approximation UNDERESTIMATES nearby true function values.

Indeterminate forms and L'Hôpital's Rule

6 Evaluate $\lim\limits_{x\to 0}\frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.
Form $\frac{0}{0}$: differentiate top and bottom: $\lim\limits_{x\to 0}\frac{\cos x}{1}=\cos 0=1$, matching the well-known limit.

7 Evaluate $\lim\limits_{x\to 0}\frac{e^x-1-x}{x^2}$ using L'Hôpital's Rule (applied twice).
Form $\frac{0}{0}$: $\lim\limits_{x\to 0}\frac{e^x-1}{2x}$, still $\frac{0}{0}$: apply again: $\lim\limits_{x\to 0}\frac{e^x}{2}=\frac{1}{2}$.

8 Evaluate $\lim\limits_{x\to\infty}\frac{\ln x}{x}$ using L'Hôpital's Rule.
Form $\frac{\infty}{\infty}$: $\lim\limits_{x\to\infty}\frac{1/x}{1}=\lim\limits_{x\to\infty}\frac{1}{x}=0$.

Recognizing when L'Hôpital's Rule applies

9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x\to 0}\frac{x+1}{x}$, and evaluate the limit by other means.

This is NOT an indeterminate form — as $x \to 0$, the numerator approaches 1 (not 0) while the denominator approaches 0, so it's a $\frac{1}{0}$ form, not $\frac{0}{0}$. This limit does not exist (it approaches $\pm\infty$ depending on direction), so L'Hôpital's Rule doesn't apply.

Differentials

10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

$dy = f'(x)\,dx = 3x^2\,dx = 3(4)(0.1) = 1.2$. Actual: $f(2.1) = 9.261$, $f(2) = 8$, $\Delta y = 1.261$. The differential $dy = 1.2$ closely approximates the actual change $\Delta y = 1.261$.

More L'Hôpital's Rule practice

11 Evaluate $\lim_{x\to 0}\frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).

Form $\frac{0}{0}$: $\lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2}$, still $\frac{0}{0}$: apply again: $\lim_{x\to 0}\frac{2\sec^2 x\tan x}{6x}$, still $\frac{0}{0}$: apply once more: $\lim_{x\to 0}\frac{2\sec^2 x(\sec^2 x) + 4\sec^2 x\tan^2 x}{6} = \frac{2(1)+0}{6} = \frac{1}{3}$.

Visualizing a linear approximation

12 The graph shows $f(x) = \sqrt{x}$ together with its tangent line at $x = 4$, $L(x) = 2 + \frac{1}{4}(x - 4)$.

The tangent line closely tracks $f(x) = \sqrt{x}$ near $x = 4$ but lies above the curve further away, visually confirming that the linearization overestimates f (since $\sqrt{x}$ is concave down) — consistent with the earlier conclusion.

L'Hôpital's Rule with exponential indeterminate forms

13 Evaluate $\lim_{x\to\infty}\frac{x^2}{e^x}$ using L'Hôpital's Rule (applied twice).

Form $\frac{\infty}{\infty}$: $\lim_{x\to\infty}\frac{2x}{e^x}$, still $\frac{\infty}{\infty}$: apply again: $\lim_{x\to\infty}\frac{2}{e^x} = 0$.

Linear approximation in an applied context

14 The side length of a cube is measured as 5 cm, with a possible measurement error of ± 0.02 cm. Use the differential $dV = 3s^2\,ds$ to estimate the maximum resulting error in the computed volume $V = s^3$.

$dV = 3(5)^2(0.02) = 3(25)(0.02) = 1.5$ cm^3. The volume could be off by as much as approximately 1.5 cm^3.

A capstone L'Hôpital's Rule problem

15 Evaluate $\lim_{x \to 0^+} x \ln x$ by first rewriting it as a quotient $\frac{\ln x}{1/x}$ to apply L'Hôpital's Rule.
Rewritten as $\frac{\ln x}{1/x}$, this is $\frac{-\infty}{\infty}$ as $x \to 0^+$: $\lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0$.

16 A surveyor measures the angle of elevation to a tower as $\theta = 60°$ with a possible error of $\pm 1°$ ($\pm \frac{\pi}{180}$ radians). If the tower's height is estimated using $h = 100 \tan \theta$ (with the base 100 m away), use the differential $dh = 100 \sec^2 \theta \, d\theta$ to estimate the resulting error in the height, to two decimal places.

At $\theta = 60° = \frac{\pi}{3}$: $\sec^2 \theta = 4$. $dh = 100(4)\left(\frac{\pi}{180}\right) \approx 400(0.01745) \approx 6.98$ m — the height estimate could be off by roughly 7 meters.

17 Evaluate $\lim_{x \to \infty} x \sin\left(\frac{1}{x}\right)$ by rewriting it as $\frac{\sin(1/x)}{1/x}$ and applying L'Hôpital's Rule (or recognizing the standard limit form).

Let $u = \frac{1}{x} \to 0^+$ as $x \to \infty$: $\lim_{u \to 0} \frac{\sin u}{u} = 1$ (the standard limit), so $\lim_{x \to \infty} x \sin\left(\frac{1}{x}\right) = 1$.

18 For $f(x) = \cos x$ near $x = 0$, complete each step:

a Find the linearization $L(x)$ at $x = 0$. $f(0) = 1$, $f'(0) = -\sin 0 = 0$: $L(x) = 1$.

b Use $L(x)$ to estimate $\cos(0.05)$. $L(0.05) = 1$.

c Compare to the true value $\cos(0.05) \approx 0.99875$, and state whether the linearization over- or under-estimates. The true value is slightly less than 1, so the linearization (which gives exactly 1) OVERESTIMATES — consistent with $\cos x$ being concave down near $x = 0$.